

From Postures to Platform: OS Design Guidelines for Touchscreen Laptops

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Abstract

Touchscreen laptops combine the intuition of touch with the productivity of desktop workflows. However, touch remains underused in office settings. In this paper, we report a mixed-methods study involving knowledge workers using their own touchscreen laptops in realistic tasks that aims to identify key interaction patterns that constrain touch adoption. We identify a set of arm postures that have been used by the participants to fight fatigue, stabilize the screen, and switch from touch to keyboard/trackpad for precision and compatibility. We translate these findings into operating-system design guidelines that facilitate reach-aware UI placement, imprecision-tolerant controls, and gesture consistency with mobile touch conventions, without degrading non-touch input. We further illustrate the application of these guidelines through a prototype gesture-based navigation system (edge gestures, taskbar swipes, reachability mode, and movable app menus) and report early validation via analytic modeling and user-based evaluation with touchscreen users. The proposed guidelines aim to support a more fluid, learnable, and ergonomic touch experience for future laptop operating systems and applications.

CCS Concepts

• **Human-centered computing - Interaction design - Empirical studies in interaction design**; • **Human-centered computing - Human computer interaction (HCI) - Interaction devices - Touch screens**;

Keywords

Touchscreen laptops, hybrid devices, posture, reachability, gesture navigation, OS interface design, multimodal interaction, ergonomics

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1 Introduction

Current design of touchscreen laptops is positioned between two interaction modalities: on one hand, the traditional desktop OS conventions that are optimized for high-precision pointing (mouse/trackpad) and on the other hand, the more modern mobile OS conventions optimized for direct touch. This is affecting the user experience negatively, undermining the wider adoption of touch in office work environments. Our study, reported here, aims at addressing these current design limitations by defining a set of design guidelines for future touch laptop operating systems that may improve the user experience. To achieve this objective, we first observed knowledge workers who used their devices in typical office work in a mixed-methods study. The users had to coordinate keyboard, trackpad/mouse, and touchscreen, often within the same task flow. This study investigated how knowledge workers interacted with the touchscreens in a work environment, revealing various posture patterns, constrained reach zones, and recurring breakdowns (precision, wobble, inconsistent touch behavior, limited text selection, and underused gesture navigation). The findings of this study lead to a set of operating system design guidelines. These principles prescribe how future laptop OSs allow touch to become ergonomically viable, consistent with user expectations from mobile touch, and compatible with non-touch workflows. Based on these guidelines, we made the first step by implementing a prototype gesture navigation system that allowed for edge gestures, taskbar swipes, reachability mode, and movable app menus, and performed a preliminary evaluation via analytic modeling and user questionnaires.

Prior work relevant to touchscreen-laptop use in productivity settings comes mainly from broader research on touch ergonomics, hybrid-device behavior, and cross-modality interaction. A systematic review by Orphanides and Nam emphasizes that touchscreen findings depend strongly on technology, context/environment, and user population, motivating context-specific study designs for knowledge work [8]. Preference studies show touch is often chosen selectively for certain tasks rather than as a general-purpose modality [12]. Posture and biomechanics are repeatedly implicated in touch performance: smartphone posture affects accuracy [4], and tablet configuration/gesture choices influence strain and performance [5], echoing concerns about fatigue and stabilization during prolonged touch [2]. Hybrid-device research further reports frequent switching among configurations and input modes within the same task, implying that interfaces should remain usable across rapidly changing modalities [9]. Work on pen and touch proposes a functional division of labor (pen for writing/drawing;



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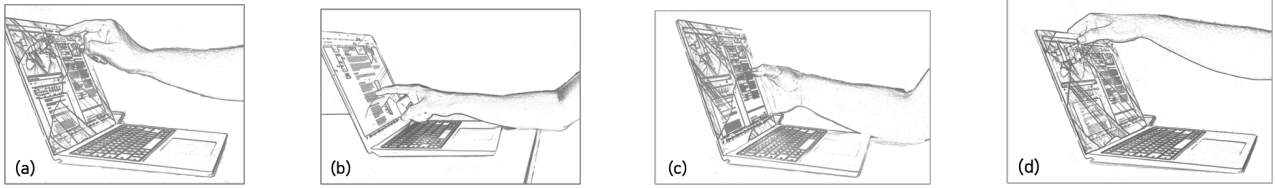


Figure 1: The four typical arm postures observed during the study: (a) Freehand touch, (b) Arm resting (c) Edge support grip (d) Top-edge support

touch for navigation/manipulation) and highlights compound interactions [3], while cognitive/experiential differences between stylus and direct touch have also been explored [11]. Finally, OS-level design history matters: Windows iterations have alternated between touch-first and mouse-first priorities, shaping today’s mixed interaction expectations [6, 10].

The research questions to address are the following: RQ1: What are the typical patterns of interaction of touchscreen laptops in office work? RQ2: What OS interaction-layer changes reduce touch friction in posture-constrained laptop use? RQ3: What early evidence suggests the proposed gesture-first design improves key navigation flows?

2 Patterns of use of touch screens

We observed nine touchscreen-laptop owners (2F/7M; mean age 30.7), who can be described as knowledge workers, a mix of students/professionals. They were asked to perform realistic office tasks on their own devices (mean 13.9”, mostly Windows 11) in their usual environments. We ran a mixed-methods, naturalistic study that was organized as follows. Each participant completed a pre-interview, two recorded task sessions (“normal use” vs “use touch as much as possible”) with think-aloud, and a post-questionnaire [7]. Tasks followed a fixed script designed to elicit multitasking (music listening while working on their own file, followed by video and written tutorial search on separate windows), while the specific apps/work content were chosen by the participants. This study has also been described and discussed in [15].

We observed two patterns of use: the *low-touch vs high-touch pattern*. During “normal use,” five participants did not use touch at all (low touch group), while four participants used it to varying degrees, often for navigation, scrolling, or specific apps. This naturally separated participants into these two profiles.

We analyzed 249 minutes of recordings and coded touch episodes into four posture classes (support, reach, comfort/stability). The four recurring arm postures (see Figure 1) were: (a) Freehand touch (full reach, high fatigue “gorilla arm”), (b) Arm resting (reduced fatigue, reach biased to lower screen), (c) Edge support grip (stabilizes wobble, reach near side edges), and (d) Top-edge support (stabilizes wobble, supports accessing top region). Their frequencies were Freehand $\geq 8/9$, Arm-resting $\geq 7/9$, Edge-support $\geq 4/9$, Top-edge $\geq 4/9$.

These postures imply reachable zones, which means that “touch-friendly OS controls” must be designed for where hands naturally are, especially near the lower screen when hands hover close to the keyboard/trackpad, and near edges when stabilizing the screen. In

Figure 2, the approximate reach areas related to the observed postures are shown. These are based on recordings analysis, highlighting an area that warrants further in-depth research. The freehand touch posture (a) allows reach of the entirety of the screen area and thus is more suitable for short actions at any part of the screen, as it is related to high fatigue. The Arm Resting Posture (b) constraints the reach area to the bottom of the screen (Figure 2(b)), an area which is even smaller if the user rests the wrist on the top of the keyboard. On the other hand, other postures constrained the reach area. Most specifically, a common pattern, also observed in mobile phones [4], is that the non-dominant hand is used for gripping the screen, with the thumb or the index finger of the dominant hand used for touch gestures. In this case, the reach area is constrained either to the side of the screen for the Edge Support Posture (c), Figure 2(c) or the top of the screen for the Top Edge Support Posture (d) (Figure 2(d)).

The study also highlighted several OS- and ecosystem-level barriers that repeatedly pushed users back to the trackpad/mouse. These included, on one hand, *precision limits* that were a combination of the “fat finger” problem and the vertical screen angle. This made reaching small targets more difficult and thus more errors were observed. A second problem was related to the “wobbling screen”. The users stabilized the display via edge/top grips that were identified earlier as typical postures. So, these postures, and in particular (c) and (d) were the typical workarounds of the users to tackle the hardware limitations and the UI design. A third problem was related to *inconsistent touch semantics*: e.g., single vs double tap expectations differ across applications and contexts, disrupting learned desktop routines. Another issue was related to weak OS-level *text selection* support: some applications offered no reliable touch text-selection, blocking “touch-only” flows in coding/editing tools. Finally, we observed a *gesture discoverability/consistency gap*: participants largely did not use OS gesture navigation, and several asked for gestures “like phones/tablets” during the post interviews. So, the design implication of this study has been that improving touch on laptops does not just need a “bigger buttons” solution, but rather requires reach-aware OS layout, imprecision-tolerant interaction primitives, and gesture conventions aligned with the user’s existing touch mental model, while keeping keyboard/trackpad workflows intact. In the next section, we will attempt to encode the findings of our study in the form of design guidelines for touchscreen Operating Systems.

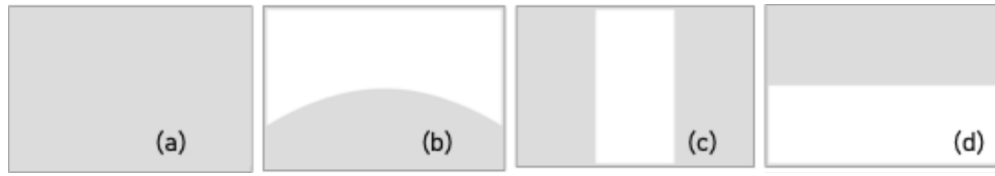


Figure 2: Screen areas that the user can approximately reach, corresponding to the typical arm postures (a) Freehand touch (full reach), (b) Arm resting, (c) Edge support grip, (d) Top-edge support

Table 1: Touch Operating System Design Guidelines

Guideline	Study Observations
G1. Design for imprecision-first touch, not precision-first UI	Touch is error-prone on desktop UIs (small targets “fat finger” plus vertical screen angle)
G2. Place high-frequency OS controls in low-effort reachable zones	Users prefer low-effort reach zones (hands near keyboard/trackpad)
G3. Support edge-based stabilization: design interactions that work with grip postures	“Wobbling screen” leads to stabilization grips (edge/top support)
G4. Make gesture navigation consistent with mobile conventions	Current gesture use is low
G5. Prefer one-finger gestures for core navigation	Multi-finger gestures are hard in laptop postures and were largely unused/unknown
G6. Add an OS-level reachability mode for the upper UI	Top UI (menus/titlebar controls) is hard to reach without fatigue
G7. Allow adaptive placement of application menus and toolbars	Posture/reach varies across users and moments
G8. Don’t break non-touch: make features work the same across input modes.	Users frequently switch between touch and keyboard/trackpad/mouse
G9. Provide a universal touch text model (selection, editing)	Current OS-level text selection/editing is inconsistent forcing the use of trackpad/keyboard.
G10. Design for micro-transitions between modalities	Micro-transitions between modalities have been observed from the initial study (one hand on the touchscreen, the other on the keyboard/trackpad)

3 Design Guidelines for future touchscreen Operating Systems

In this section, we present ten Design Guidelines that address the findings of the previous study, shown in Table 1. Then we describe their operationalization in the form of an Operating System, KotoOS, a touch-centric OS navigation and multitasking concept implemented as an interactive high-fidelity prototype. KotoOS does not attempt to redesign the full desktop modalities; instead, it targets the OS-level interaction layer that most strongly shapes everyday workflow: returning to the desktop, switching virtual desktops, accessing launcher/search, and invoking system controls/notifications. The design is grounded in two constraints observed in real laptop use, as discussed in the previous section: users often interact from low-effort postures near the keyboard and along screen edges, in line with Bragdon et al. [1] observation that touch interactions near the edge of the screen can be as fast as physical buttons. In addition, desktop UI frequently assumes high-precision pointing, which makes touch error-prone and discouraging.

The design rationale of KotoOS embodies three principles. First, it is imprecision-first: core actions should not require accurate tapping of small targets; therefore, KotoOS prioritizes gestures and large interactive regions (e.g., taskbar swipes, corner gestures).

Second, it is reach-aware: frequent actions are invoked from the bottom edge and corners, aligning with hand position near the keyboard and reducing arm elevation. Third, it leverages familiar mobile conventions to reduce learning cost, adopting patterns such as “swipe up to go home” and “swipe up and hold for multitasking.” In addition, to accommodate laptop touch postures and screen stabilization behaviors, KotoOS uses one-finger gestures for all primary actions.

KotoOS was implemented in Figma as a set of interactive scenarios that demonstrate gesture-triggered state changes: desktop, multitasking, application launcher, and system panels. Due to tool constraints, multiple interaction outcomes were authored as separate cases, enabling controlled evaluation of key flows and side-by-side comparison with existing OS behavior, see Figure 3 for examples of the prototype. The core gestures and the corresponding functions of KotoOS implementation were the following: (a) Swipe up from bottom → *Go to desktop/home*, (b) Swipe up + hold → *Multitasking view*, (c) Arc-like gesture at bottom of screen → *Switch between desktops*, (d) Bottom-left corner swipe → *App launcher (enables hybrid use of touch → launch app, with the trackpad → use app)*, (e) Bottom-right corner swipe → *Control and notifications panel*. In addition, we implemented a Reachability mode (swipe

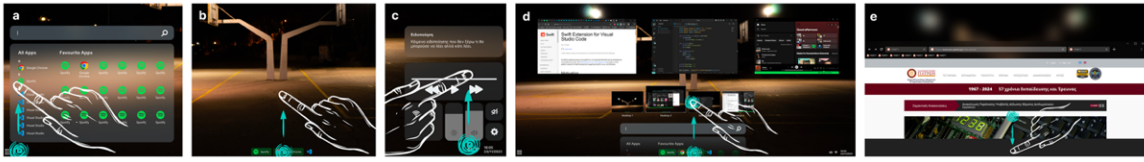


Figure 3: Instances of KotoOS prototype interaction: (a) Swipe up from the left bottom corner to go to the launcher, (b) Swipe up from the middle bottom to go to the desktop, (c) Swipe up from the right bottom corner to go to the Control Panel, (d) Swipe up from the middle bottom and hold to go to multitasking view (e) Swipe down towards the middle bottom to enable reachability

down) to bring the top UI within reach, Search (swipe down on the desktop wallpaper), and Optional movable app menus (bring toolbars closer to hands).

These key functionalities directly target study observations of Table 1 reach to top menus, lack of gesture adoption, need for imprecision-tolerance, and posture-constrained comfort zones. Next, we report on a preliminary evaluation of this prototype.

4 Preliminary Evaluation

We complemented the design with a preliminary evaluation of KotoOS using an analytic and a qualitative method. KotoOS is a prototype of a gesture navigation system emphasizing imprecision-tolerant interactions, bottom/edge accessibility, mobile-consistent conventions, and non-touch compatibility. First, we performed a model-based comparison (GLM/ KLM) for representative navigation tasks, and then we conducted an interactive interview from which we collected feedback from advanced touchscreen-laptop users, indicating the strongest perceived benefits for app/desktop switching and multitasking access.

In the first study KotoOS was compared against Windows 11 touch (buttons and, where available, built-in touch gestures) and Windows 11 cursor/trackpad using the Gesture-Level Model (GLM) [13], an adaptation of KLM [14], for touch that estimates expert execution time via operators such as mental preparation, finger movement, tap, drag, and flick. Across representative OS navigation tasks, GLM predicts that KotoOS often reduces precision-dependent steps and/or “find-and-hit” mental acts, while remaining competitive with cursor-based KLM times: For *Go to Desktop* (2.11s KotoOS vs 3.29s Windows tap), *Switch desktops* (2.11s KotoOS vs 5.23s Windows touch flow), *Search* (2.11s KotoOS vs 3.29s Windows tap). On average, KotoOS was faster than Windows, between 35% and 59% in all tasks, except for accessing the Notification and Control Panel, see Figure 4.

Second, we ran an interactive interview in which three advanced, touchscreen users assessed each feature. For every feature, participants first demonstrated how they currently perform the task, then tried the corresponding method(s) in Windows 11 and in KotoOS. They explained what they liked and disliked about each approach and ultimately selected a favorite. Overall, the findings indicate a strong preference for KotoOS on the most navigation-critical features—especially *Go to Desktop*, *Multitasking view*, and *Switch between desktops*, often credited to its one-finger, mobile-familiar interaction style that felt more predictable and less error-prone. Other features were judged as comparable (e.g., the *launcher*) or produced mixed feedback: *Search* showed little perceived difference;

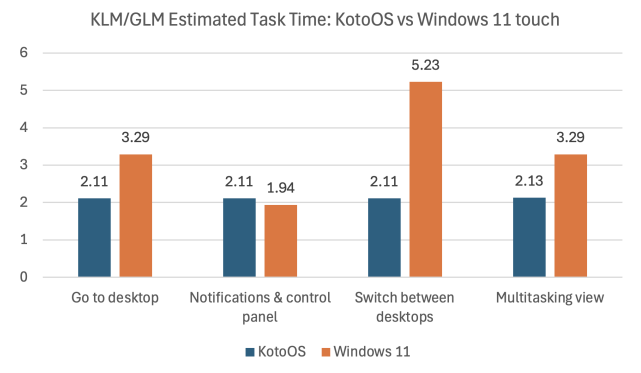


Figure 4: Approximate execution time of key tasks, measured through KLM/GLM (lower is better)

the *Control/Notifications panel* was faster in Windows, but KotoOS was favored for merging the two panels; *Reachability* was viewed positively, though slightly less predictable; and *Movable menus* were appreciated, although one participant preferred conventional top menus.

5 Conclusion and Future work

The reported work attempted to answer the key questions that relate to identifying and obtaining early signals of core design guidelines of a touchscreen laptop Operating System that addressed observed limitations of current OS design.

The findings of our naturalistic study suggest that touchscreen laptops are currently “hybrid by hardware but not by interaction”. Participants frequently fell back to the trackpad/mouse even when touch felt natural, because the desktop interaction layer remains optimized for precision pointing rather than imprecise, posture-constrained touch. In our study with nine touchscreen-laptop owners, five did not use touch at all during normal work, while four used it opportunistically, indicating that “touch-capable” does not automatically translate to “touch-adopted” in office workflows. The observed breakdowns (precision limits, screen wobble, inconsistent touch semantics, weak touch text-selection support, and poor gesture discoverability/consistency) show that friction is not only a UI size problem, but an OS interaction design problem.

The proposed guidelines and their operationalization attempt to address these issues. The preliminary evaluation of the KotoOS prototype was performed using a combination of an analytic

expert-model comparison and a user-based evaluation. Analytic modeling (GLM vs KLM) indicates that several navigation tasks can be executed with fewer mental/targeting costs using gesture-first, bottom-access interactions (e.g., “Go to desktop,” “Switch between desktops,” “Go to multitasking”), while users showed preference for most of the new functionality, in comparison to the corresponding Windows 11, that was mostly multi-finger swipes.

This work is intentionally exploratory and has several limitations. First, the observational component involved a small sample ($n=9$) of mostly young knowledge workers using predominantly Windows 11 touchscreen laptops, which limits generalizability across age groups, occupations, device sizes/form factors, and operating systems. Second, the study tasks were designed to elicit multitasking and realistic office behaviors but still represent a narrow slice of knowledge work; other domains (creative work, intensive spreadsheet use, accessibility contexts) may yield different posture strategies and interaction breakdowns. Third, KotoOS is a high-fidelity interactive prototype (Figma) rather than an instrumented OS implementation, which constrains what can be measured (e.g., true motor performance, error rates, unintended activations, discoverability in the wild, and real multitasking latency). The user-based evaluation involved only three touchscreen users and focused on subjective judgments per feature; this is appropriate for early design feedback but does not establish population-level effects or long-term adoption, limitations that make it difficult to assess whether the proposed OS changes would reliably improve productivity in realistic hybrid workflows.

A key contribution of this work is the posture-driven framing of touch interaction on laptops. The four recurring postures (freehand, arm resting, edge support grip, top-edge support) help explain why common desktop placements (top menus, small window controls) and two-handed/multi-finger gestures are ergonomically brittle on laptops. These postures imply reach-biased “comfort zones” and stabilization behaviors that OS design can either fight or exploit. This directly motivated our guidelines emphasizing reach-aware placement (e.g., bottom/edge invocation), imprecision-tolerant controls, and mobile-consistent, one-finger gesture conventions that remain compatible with non-touch input.

The KotoOS prototype demonstrates how multiple guidelines can be operationalized in a coherent OS navigation layer: bottom-edge “home/desktop,” swipe-up-and-hold multitasking, swipe-desktop switching, a consolidated control/notification panel, reachability mode, and optional movable app menus. Our early validation suggests that benefits are not uniform across OS functions: model-based estimates indicate the strongest gains when gestures replace precision-dependent “find-and-hit” actions (e.g., desktop/app/desktop switching), while user feedback indicates some functions may feel comparable to existing OS behavior (e.g., launcher/search) and others create new trade-offs (e.g., reachability predictability; preferences around movable menus). This points to an important design implication: gesture-first design is most valuable when it reduces precision demand, hand travel, and cognitive overhead—without introducing mode errors or harming expert keyboard/trackpad flows.

Finally, our results highlight a broader ecosystem challenge: even if OS navigation becomes touch-friendly, touch adoption will remain constrained if core productivity tasks (notably text selection/editing across applications) are not supported by a consistent,

OS-level “touch text model.” This motivates guideline-level interventions that go beyond navigation gestures toward system services and cross-app consistency, arguably the missing “platform layer” for laptop touch.

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